

Fidel González Leiva

Robotics and Automation Engineer | Software Developer

Jerez de la Fra.

11408

Cádiz

+34 601 034 019

lotape6@gmail.com

in Fidel González Leiva

lotape6

I usually go wrong, but every time I make better mistakes

Professional Summary

I am an empathetic and passionate software developer with a deep love for technology and a strong drive for teamwork. With excellent leadership skills and assertiveness, I thrive in collaborative environments, consistently delivering innovative solutions while fostering a positive and productive team culture.

Education

- 2014–2017 **College degree**, UMA, Málaga, *Electronic Robotic and Mechatronic Engineering*
- 2017–2018 **College degree**, US, Sevilla, *Electronic Robotic and Mechatronic Engineering*
- 2018–2019 **Master's degree**, US, Sevilla, *Electronic, Robotics and Automation Engineering*

Experience

July 2022 - **Software Developer**, Bertrandt Group

- May 2024
 - Developed automotive-critical software using AUTOSAR and C++.
 - Analyzed, implemented, and tracked low-level requirements.
 - Extensively used git for the integration of highly dependent developments.
 - Identified and resolved bugs through analysis of CAN, LIN, SOME/IP, and DLT traces.
 - Worked under Agile methodologies (SCRUM)
 - *Technologies*: C++, GTest, GMock, AUTOSAR, Python, GDB Debugger Docker (under WSL), CMake, Bash and Wireshark.

June 2019 - **Robotics Engineer | Software Developer**, FADA-CATEC

- July 2022
 - **MBZIRC 2020**: Designed computer vision algorithms for aerial target interception.
 - **GAVIOTA**: Developed ground-based AI systems for aircraft detection.
 - **DELOREAN**: Collected and analyzed data for autonomous UAV operations in urban environments.
 - **METRICS**: Organized and evaluated an international championship on UAV navigation and defect detection.
 - **Piloto5GAndalucia**: Developed a drone swarm for real-time security tasks.
 - **ASSISTANCE**: Led a team developing multiple UAVs for disaster relief.
 - *Technologies*: C++, Python, ROS, OpenCV, Gstreamer, GDB Debugger, CMake, CUDA, GMock, GTest, Docker, Bash, embedded Linux distributions, image labeling tools and $\LaTeX$.

Personal Projects & Independent Learning

September 2024 - **Cybersecurity**, Read Team | Blue Team

- January 2025
 - **Pentesting** Practicing different tactics, techniques, and procedures to gain access and escalate privileges in different servers (mainly Windows and Linux machines).
 - *Commonly exploited protocols*: HTTP, SMB, LDAP, FTP, SSH, RPC, NFS.
 - *Web exploits*: Web content discovery, DNS discovery, SQL injection, bypassing of login forms, SSTI, XSS, IDOR, LFI/RFI, RCE.
 - *Others techniques*: Active directory enumeration (SharpHound + BloodHound), privilege escalation, container escaping, password cracking, brute-forcing, payload obfuscation.

- **SOC Analyst** Learning about cybersecurity principles, threat hunting, incidence response, and malware reversing.
 - Hands-On experience with malware static/dynamic analysis, decodification/deobfuscation of payloads, memory forensics, and OpenSource EDR solutions.
 - *Tools*: Wazuh, Volatility, CyberChef, SpeakEasy, Sysmon, Ghidra, GDB, and basic command line tools combination (strings, grep, sed, xargs, etc.)

September **Artificial Intelligence, NLP/LLM Models**

- 2024 – Studied LLM models, tokenizers, and Retrieval-Augmented Generation concepts.
- November 2024 – Implemented a small local LLM AI agent using RAG with PGVector to enhance the performance of the model on personal data.
- *Tools*: LlamaIndex, Llama-Cpp, Python, PostgreSQL, PGVector.

Languages

Spanish Native

English Cambridge B2 First Certificate *(last 3 years working mostly in English)*

Certifications and Courses

2025	EDR: Threat Hunting & Threat Intelligence	Udemy
2023	Introduction to the AUTOSAR Classic Platform	dSpace
2022	Nanodegree in C++	Udacity
2021	Docker Mastery: With Kubernetes + Swarm from a Docker Captain	Udemy

Interests

Playing flamenco guitar, cyber security, lock-picking and freestyle FPV drones

Publications

- [1] Manuel García, Rafael Caballero, Fidel González, Antidio Viguria, and Aníbal Ollero. Autonomous drone with ability to track and capture an aerial target. In *2020 International Conference on Unmanned Aircraft Systems (ICUAS)*, pages 32–40, 2020.
- [2] Fidel González, Rafael Caballero, Francisco J. Pérez-Grau, and Antidio Viguria. Vision-based uav detection for air-to-air neutralization. In *2021 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, pages 236–241, 2021.
- [3] Jorge Mariscal-Harana, Víctor Alarcón, Fidel González, Juan José Calvente, Francisco Javier Pérez-Grau, Antidio Viguria, and Aníbal Ollero. Audio-based aircraft detection system for safe rpas bvlos operations. *Electronics*, 9(12), 2020.

Timeline of personal interest in technologies

